

Petrochemical Technology

Polymers

Polymers

A polymer is a long chain molecule that is composed of a large number of repeating units of identical structure. Polymers may be naturally occurring like proteins, cellulose and silk or synthetic like polyethylene, polystyrene and nylon. In some cases, naturally occurring polymers can also be produced synthetically. An important example is natural rubber, which is known as polyisoprene in its synthetic form.

Polymers that are capable of high extension under ambient conditions find important applications as elastomers. In addition to natural rubber, there are synthetic elastomers such as nitrile and butyl rubber. Other polymers may have characteristics that permit their formation into long fibers suitable for textile applications. The synthetic fibers, principally nylon and polyester, are good substitutes for the naturally occurring fibers such as cotton, wool and silk.

In contrast to the usage of the word polymer, those commercial materials, other than elastomers and fibers that are derived from synthetic polymers are called **plastics. A commercial plastic resin may contain two or more polymers in addition to various additives and fillers. These are added to improve some property such as the processibility, thermal or environmental stability and mechanical properties of the final product.**

Today, polymeric materials are used in nearly all areas of daily life and their production and fabrication are major worldwide industries. As a group, plastics accounted for over 75% of their production, while synthetic fibers (15%) and elastomers (10%) made up the rest.

Polymers are characterised into four classes:

Thermoplastics consist of individual molecules with no covalent bonding between them but held together by intermolecular bonding. The polymers become soft on warming and can be moulded. They can be repeatedly warmed, softened and remoulded.

Thermosets, on the other hand, have many covalent bonds between the chains, leading to a three-dimensional structure, which can be regarded as a single molecule. They can be moulded by heat and pressure, but once moulded they cannot be remoulded. The most important examples include the plastics made from methanal (formaldehyde).

Elastomers are amorphous solids which, as the name suggests, are elastic. They have coiled chains which can be stretched out but spring back to their original shape when the stretching force is released.

Fibres are thin threads which are produced by extruding a molten polymer through a die in which there are small holes. Fibres produced in this way include the polyamides (such as nylon), the polyesters (such as terylene) and poly(propene)

The major differences between smaller molecules and polymers lie not with their chemical properties but with their physical ones. Their larger sizes lead to much stronger intermolecular forces leading in turn to much higher melting points, and the characteristic properties of hardness and flexibility. These intermolecular forces are even stronger when the polymer chains pack together in a regular way as in HPDE (high density polyethylene) and have regions of crystallinity. When heated, it melts and the crystallinity is lost. As it does not have a sharp melting point, the temperature at which this occurs is termed the *melt transition temperature*, T_m . Above this temperature, the polymer is amorphous. Some polymers are hard and amorphous, having no regions of crystallinity, for example, poly(methyl methacrylate). The temperature at which they become soft and pliable is termed the *glass transition temperature*, T_g .

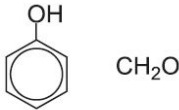
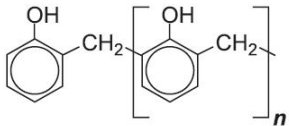
Addition Polymers

In **addition polymerization**, the polymer has the same empirical formula as the monomer but a higher molecular mass

Monomer	Formula	Polymer	Trivial name	Structure
Ethylene	$\text{H}_2\text{C}=\text{CH}_2$	LDPE <u>Low density poly(ethene)</u>	low density polythene	$-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-$
Vinyl chloride	$\text{H}_2\text{C}=\text{CHCl}$	<u>Poly(chloroethene)</u>	polyvinyl chlorine, PVC	$-\text{CH}_2-\overset{\text{Cl}}{\underset{ }{\text{CH}}}-\text{CH}_2-\overset{\text{Cl}}{\underset{ }{\text{CH}}}-$
Propylene	$\text{H}_2\text{C}=\text{CH}-\text{CH}_3$	<u>Poly(propene)</u>	polypropylene	$-\text{CH}_2-\overset{\text{CH}_3}{\underset{ }{\text{CH}}}-\text{CH}_2-\overset{\text{CH}_3}{\underset{ }{\text{CH}}}-$
Acrylonitrile	$\text{H}_2\text{C}=\text{CH}-\text{CN}$	<u>Poly(propenenitrile)</u>	polyacrylonitrile	$-\text{CH}_2-\overset{\text{CN}}{\underset{ }{\text{CH}}}-\text{CH}_2-\overset{\text{CN}}{\underset{ }{\text{CH}}}-$
Methyl methacrylate	$\text{H}_2\text{C}=\overset{\text{CO}_2\text{CH}_3}{\underset{ }{\text{C}}}-\text{CH}_3$	<u>Poly(methyl 2-methylpropenoate)</u>	polymethyl methacrylate	$-\text{CH}_2-\overset{\text{CO}_2\text{CH}_3}{\underset{\text{CH}_3}{ }{\text{C}}}-\text{CH}_2-\overset{\text{CO}_2\text{CH}_3}{\underset{\text{CH}_3}{ }{\text{C}}}-$
Styrene	$\text{H}_2\text{C}=\text{CH}-\text{C}_6\text{H}_5$	<u>Poly(phenylethene)</u>	polystyrene	$-\text{CH}_2-\overset{\text{C}_6\text{H}_5}{\underset{ }{\text{CH}}}-\text{CH}_2-\overset{\text{C}_6\text{H}_5}{\underset{ }{\text{CH}}}-$
Tetrafluoroethene	$\text{F}_2\text{C}=\text{CF}_2$	<u>Poly(tetrafluoroethene) (PTFE)</u>	Polytetrafluoroethylene (PTFE)	$-\text{CF}_2-\text{CF}_2-\text{CF}_2-\text{CF}_2-$

Condensation Polymers

In **condensation polymerization**, polymerization of one or more monomers is accompanied by the elimination of small molecules (such as water or ammonia).

Polymer	Monomer	Formula
<u>Polyesters</u>	$\text{HO}-(\text{CH}_2)_x-\text{C} \begin{array}{l} \text{O} \\ \parallel \\ \text{OH} \end{array}$	$\left[(\text{CH}_2)_x-\text{C} \begin{array}{l} \text{O} \\ \parallel \\ \text{O} \end{array} \right]_n$
<u>Polyamides</u>	$\begin{array}{c} \text{H} \quad \quad \quad \text{H} \\ \quad \quad \quad \\ \text{N}-(\text{CH}_2)_x-\text{N} \\ \quad \quad \quad \\ \text{H} \quad \quad \quad \text{H} \\ \text{O} \quad \quad \quad \text{O} \\ \parallel \quad \quad \parallel \\ \text{HO}-\text{C}-(\text{CH}_2)_y-\text{C}-\text{OH} \end{array}$	$\left[\text{NH}-(\text{CH}_2)_x-\text{NH}-\text{C} \begin{array}{l} \text{O} \\ \parallel \end{array} (\text{CH}_2)_y-\text{C} \begin{array}{l} \text{O} \\ \parallel \end{array} \right]_n$
<u>Phenol-methanal plastics</u>		
<u>Polyurethanes</u>	$\begin{array}{l} \text{HO}-\text{R}^1-\text{OH} \\ \text{O}=\text{C}=\text{N}-\text{R}^2-\text{N}=\text{C}=\text{O} \end{array}$	$\left[\text{R}^1-\text{O}-\text{C} \begin{array}{l} \text{O} \\ \parallel \end{array} \text{NH}-\text{R}^2-\text{NH}-\text{C} \begin{array}{l} \text{O} \\ \parallel \end{array} \text{O} \right]_n$

Polymerization Procedures

Polymerizations are carried out by four different procedures – bulk, solution, suspension and emulsion polymerizations.

In bulk polymerization, the monomer and the initiator are combined in a vessel and heated to the proper temperatures. This procedure, although the simplest, is not always the best. The polymer that forms may dissolve in the monomer to give a viscous mass and heat transfer becomes difficult. Heat cannot escape, and the polymer may char or develop voids. If the exotherm gets out of hand, the system may explode.

Even so the polymerization of ethylene by high-pressure method is a bulk polymerization and is one of the polymerizations carried out on the largest scale. Fortunately, the polymer does not dissolve in the monomer. Instead it collects at the bottom of the reactor and is drawn off. The exotherm still presents a problem, and the strictest possible control of temperature and heat transfer is necessary. The polymerization of methyl methacrylate to “Lucite” (Plexiglass, Perspex) is also carried out in bulk.

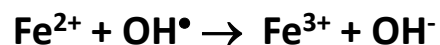
In **solution polymerization**, the reaction is carried out in a solvent that acts as a heat sink and reduces the viscosity of the reaction mixture. The snags with the solution polymerization are first, it is frequently difficult to remove the last traces of solvent from the polymer, and second, the solvent participates in chain transfer reactions so that low molecular weight polymers result. Solution polymerization is useful if the product is to be used in solvent. Solvent-based poly (vinyl acetate) adhesive is an example.

Slurry polymerization is a variant of solution polymerization and is used for the important polymerizations of ethylene to high-density (low-pressure) polyethylene, and propylene to polypropylene. A small amount of solvent is combined with the monomer and the catalyst in the reactor. The solvent forms slurry with the catalyst and aids in its distribution throughout the reaction mixture. At the same time, it helps to remove exotherm. Initially, it was necessary to separate the catalyst by a cumbersome process. This has led to the development of catalysts that separate more readily. Also, catalysts are now available that give such high yields, and thus are present in low concentrations, that they can be left in the polymer without affecting its properties.

In **suspension polymerization, the monomer and catalysts are suspended as droplets in a continuous phase such as water. These droplets have a high surface/volume ratio so heat transfer to the water is rapid. The droplets are maintained in suspension by continuous agitation and also, if necessary, by addition of a water-soluble polymer such as methyl cellulose that increases viscosity of the water. Finely divided inorganic materials such as clay, talc, aluminum oxide and magnesium carbonate have a similar stabilizing effect on the suspension. The need to remove these materials is one of the disadvantages associated with their use. Poly (vinyl chloride) is frequently made by suspension polymerization.**

Emulsion polymerization uses an emulsifying agent, usually various kinds of soap. In solution, this forms micelles in which the nonpolar hydrophobic ends of the soap molecules point inward and the polar hydrophilic groups point outward and interact with the water. If monomer is added, that can be absorbed, in the micelles, a separate monomer droplet phase may form that is also stabilized by the soap molecules, the droplets being a micron or more in diameter.

A water-soluble composite initiator called a “redox” catalyst is then added. This consists of a mixture of a reducing agent and an oxidizing agent. An example is the ferrous ammonium sulfate and hydrogen peroxide. In the absence of monomer, the former would reduce the latter in a two-stage process:



If monomer is present, however, the hydroxyl free radical can initiate polymerization.

These polymerizations must be carried out with rigorous exclusion of oxygen, which is an inhibitor in this case, although an initiator in low-density polyethylene production.

There are two important differences between emulsion and suspension polymerization. In emulsion polymerization, the catalyst or the initiator is in the aqueous phase, not dissolved in the monomer. Also the particles produced are at least an order of magnitude smaller than those obtained from suspension polymerization.

Molecular Weight Distribution

A typical synthetic polymer sample contains chains with a wide distribution of chain lengths. This distribution is seldom symmetric and contains some molecules of very high molecular weight. The exact breadth of the molecular weight distribution depends upon the specific conditions of polymerization. For example, polymerization of some polyolefins results in a molecular-weight distribution that is extremely broad, while it is possible to polymerize some polymers, such as polystyrene with nearly monodisperse distribution under laboratory conditions. Therefore, it is necessary to define an average molecular weight to characterize an individual polymer sample as detailed below.

The number average molecular weight ($\bar{M}_n$) is obtained by adding the molecular weights of all the molecules and dividing by the number of molecules. If we have n_1 molecules of molecular weight M_1 , n_2 of molecular weight M_2 and n_x of molecular weight M_x , then

$$\bar{M}_n = \frac{n_1 M_1 + n_2 M_2 + \dots + n_x M_x + \dots}{n_1 + n_2 + \dots + n_x + \dots} \quad \text{or,} \quad \bar{M}_n = \frac{\sum_{i=1}^N n_i M_i}{\sum_{i=1}^N n_i}$$

The weight average molecular weight ($\bar{M}_w$) is calculated according to weight of all the molecules at each molecular weight. Let w_1 be the weight (in molecular weight units) of molecules of molecular weight M_1 , w_2 the weight of molecules of molecular weight M_2 and so on, then

$$\bar{M}_w = \frac{w_1 M_1 + w_2 M_2 + \dots + w_x M_x + \dots}{w_1 + w_2 + \dots + w_x + \dots}$$

$\bar{M}_n$ tells us where most of the polymer molecules are relative to the molecular weight distribution. The parameter $\bar{M}_w$, on the other hand, tells us where most of the weight is regardless of the molecular weight distribution. Because $\bar{M}_w$ is biased towards molecules with higher molecular weight, it will be larger than $\bar{M}_n$

A measure of the breadth of the molecular weight distribution is given by the ratios of molecular weight averages. For this purpose, the most commonly used ratio is $\bar{M}_w/\bar{M}_n$, which is called the polydispersity index or PDI.

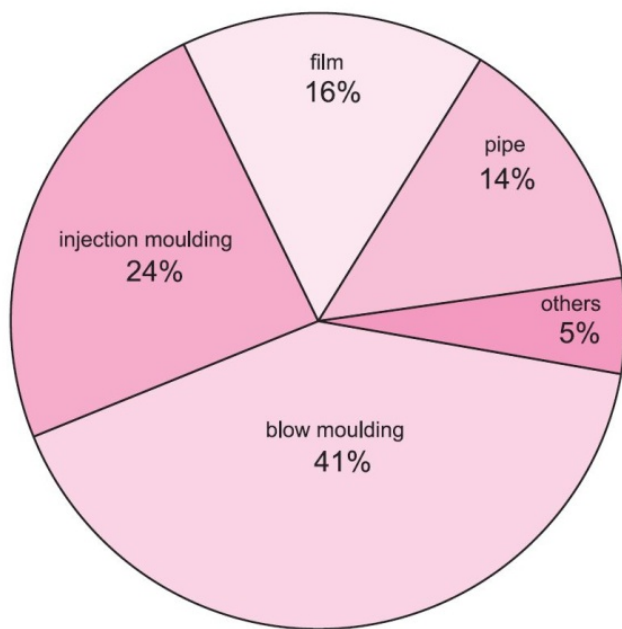
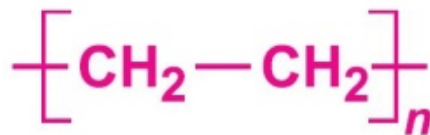
If PDI is unity then all the molecules have the same molecular weight, and as the distribution of molecular weights becomes wider, this ratio increases.

Degree of Polymerization

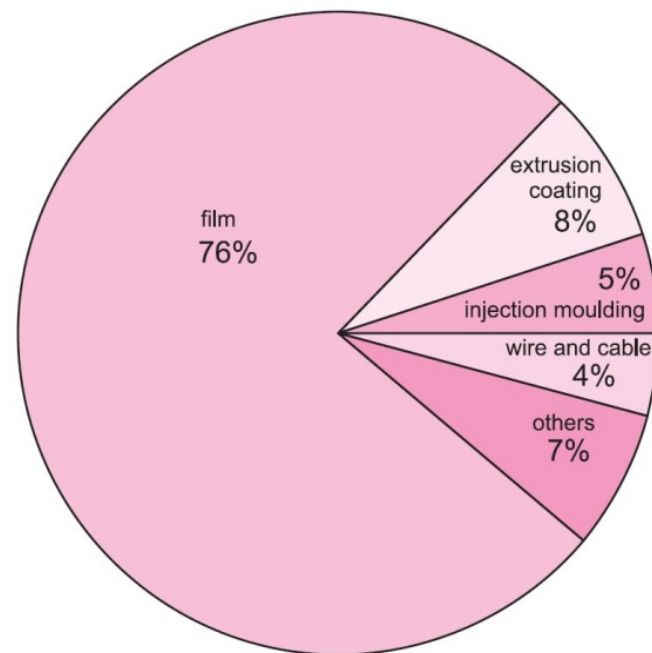
The degree of polymerization (DP) is the number of repeat units in an average polymer chain at time t in a polymerization reaction. The length is in monomer units. The degree of polymerization is a measure of molecular weight. For most industrial purposes, lengths in the thousands or tens of thousands are desired.

$$\text{DP} = \text{Total MW of the polymer} / \text{MW of the repeating unit}$$

Polyethylene (Polythene)



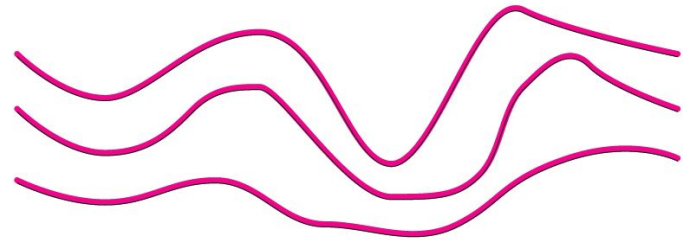
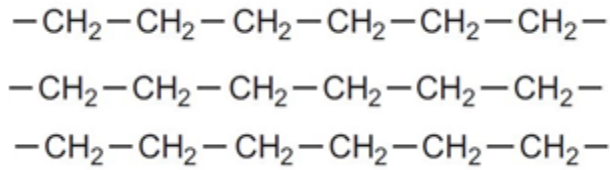
HDPE



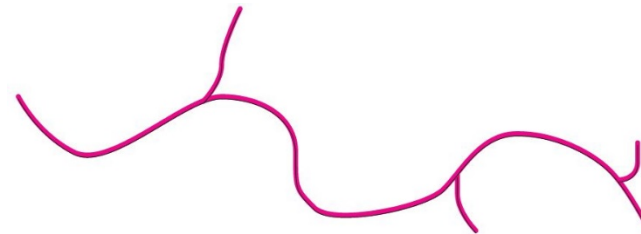
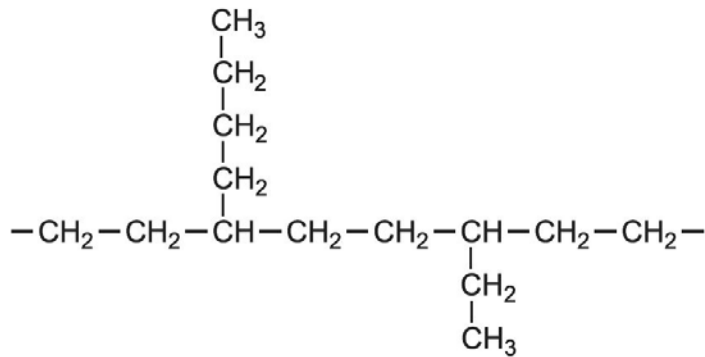
LDPE/LLDPE

Annual production: 109 million tonnes
(HDPE: 47; LDPE 24; LLDPE 28)

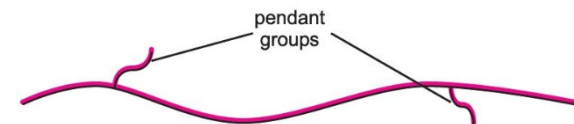
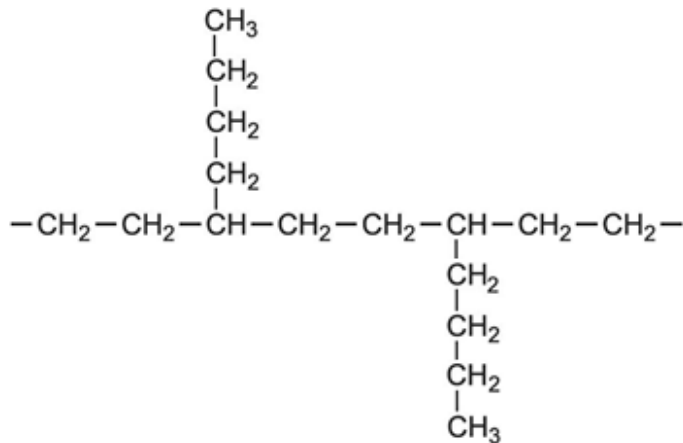
Types of Polyethylene



HDPE



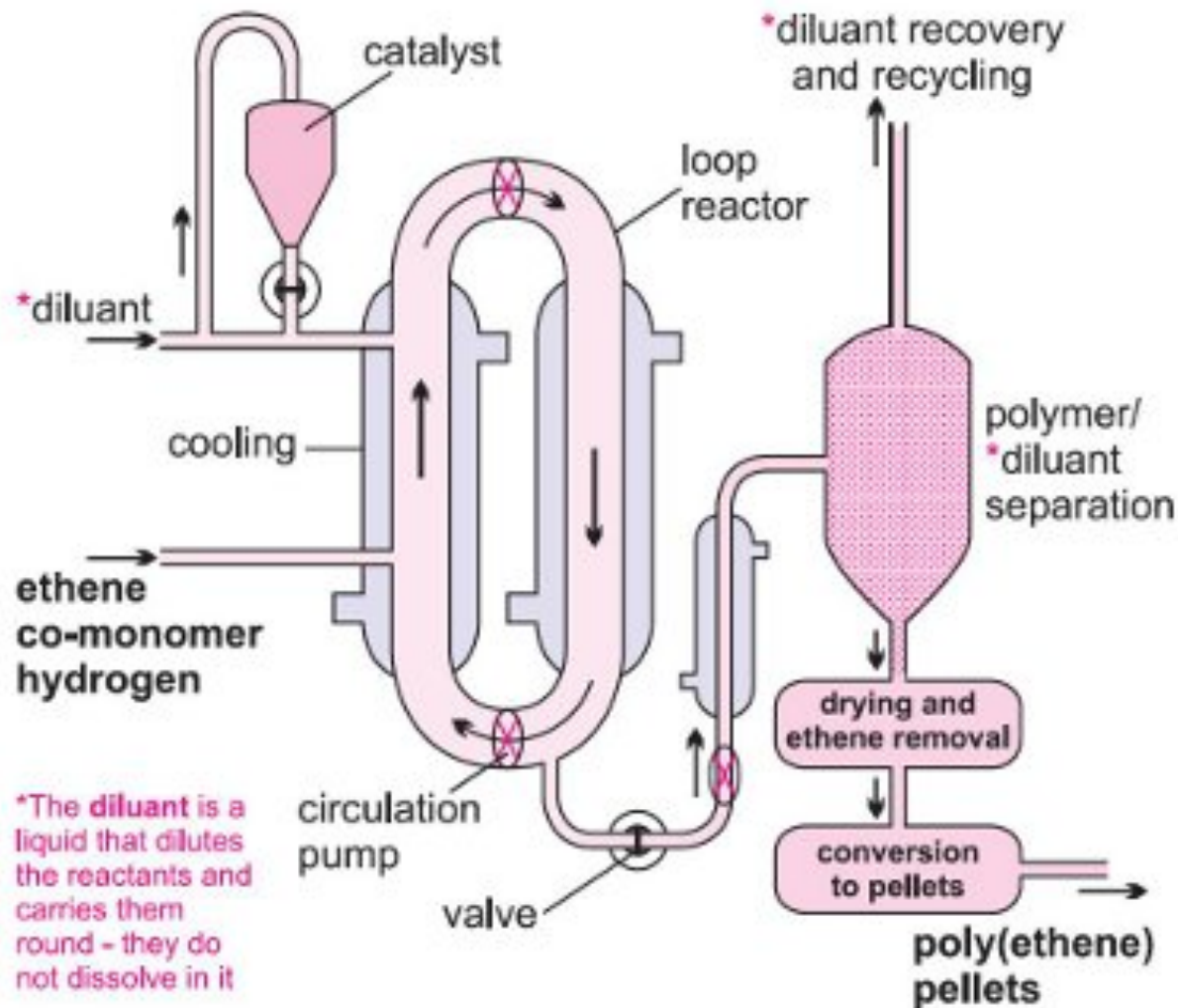
LDPE



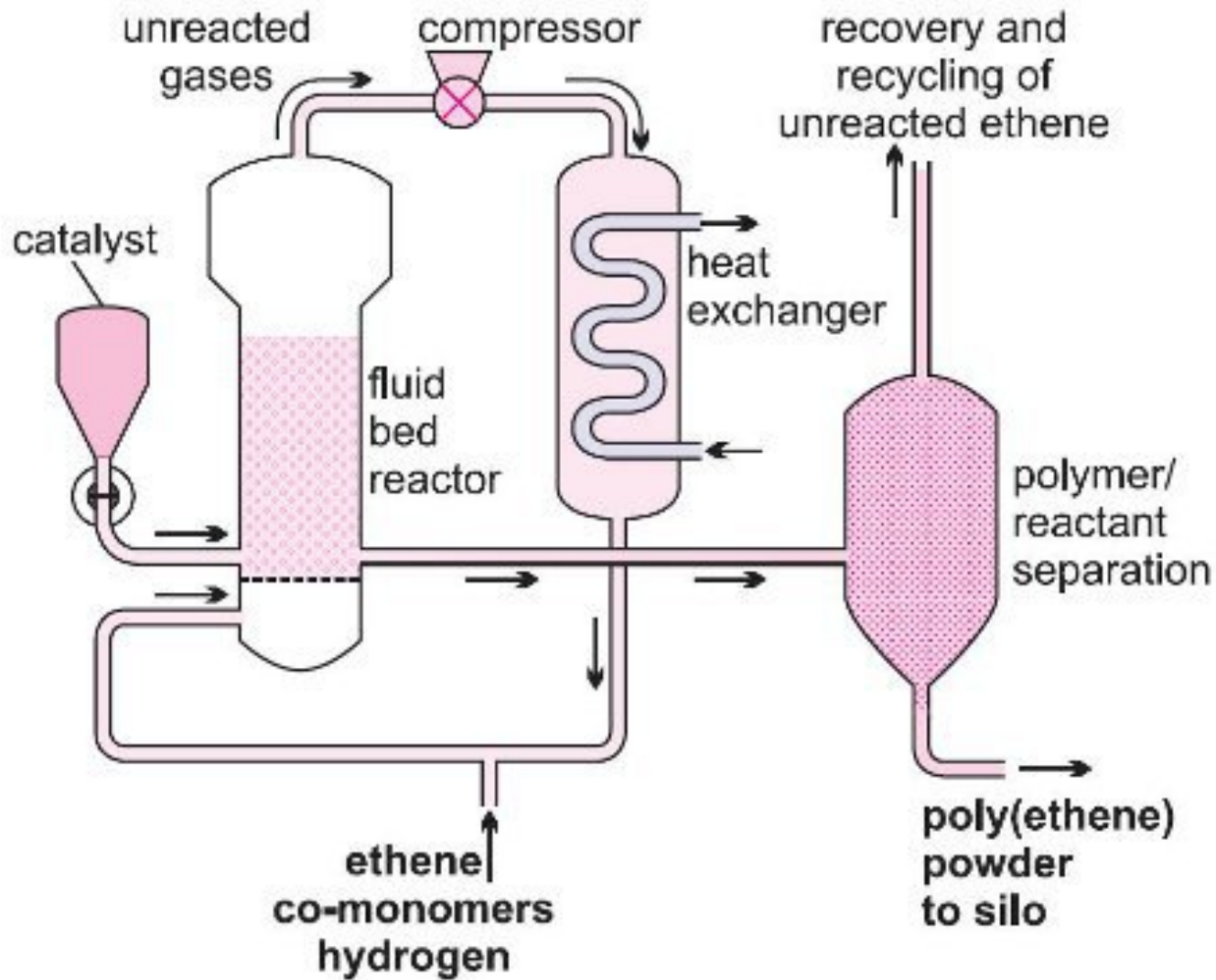
LLDPE

Properties of Polyethylenes

Parameter	LDPE	HDPE	LLDPE
Initiator or catalyst	Oxygen or organic peroxide	Ziegler or Phillips catalyst	Ziegler or Phillips catalyst
Reaction temperature (°C)	200-300°C	As low as 60°C	As low as 60°C
Pressure (Bar)	1300-2600	1-300	1-300
Structure	Branched	Linear	Linear with short branches
Approximate crystallinity	55%	85-95%	55%
Comonomer	None	None	1-Butene, 1-hexene Or, 1-octene
Tensile strength (psi)	1200-2000	3000-5500	2000-2500
Elongation at break (%)	500	10-1000	500
Density (g/cm ³)	0.915-0.925	0.945-0.965	0.915-0.925

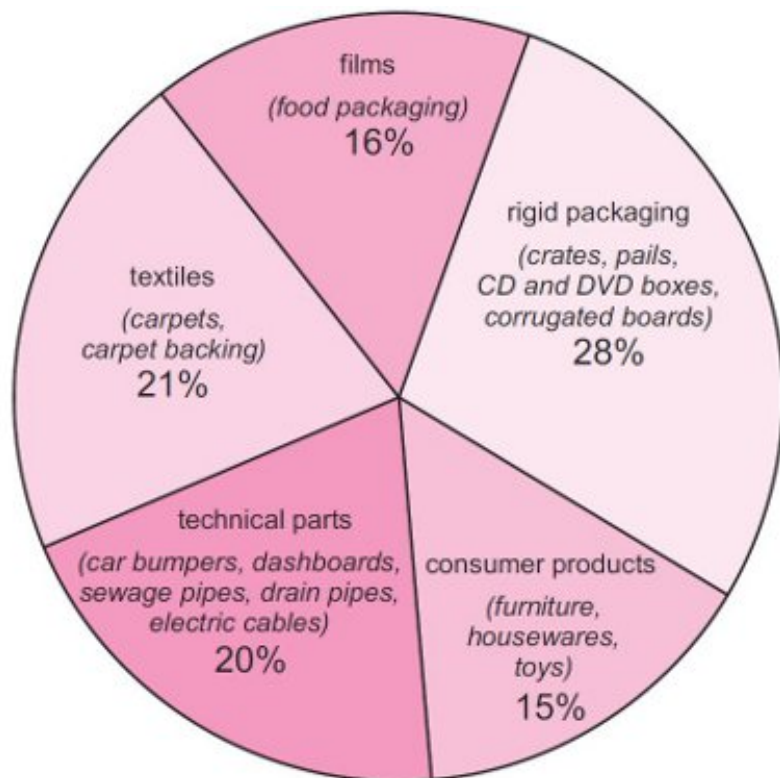
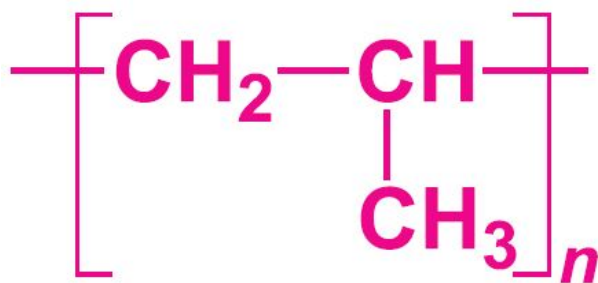


Manufacture of poly(ethylene) using slurry process in a loop reactor

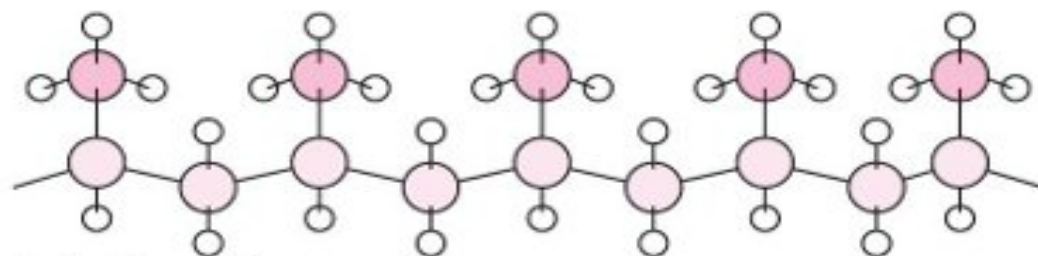


Manufacture of poly(ethylene) using Low-pressure gas-phase process

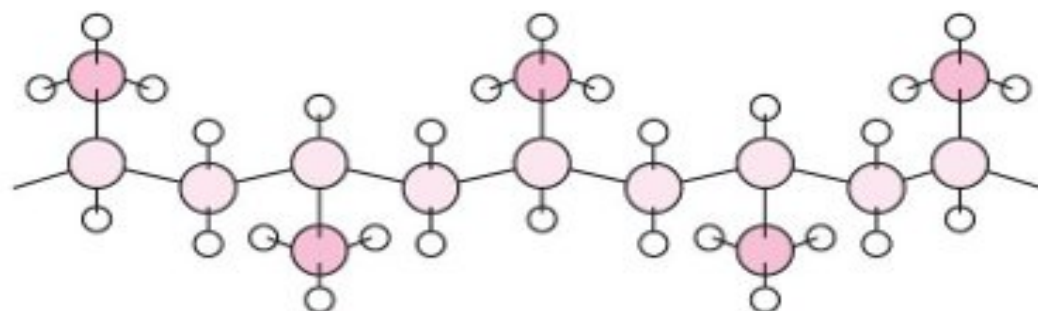
Poly(propene) (Polypropylene)



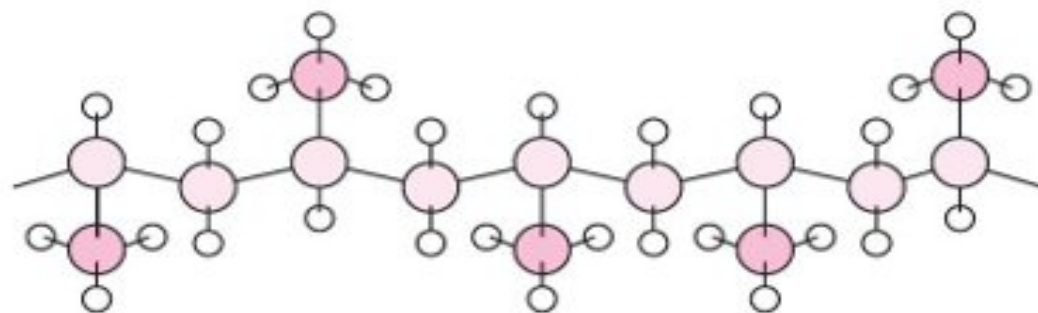
Annual Production: 95 million tonnes



isotactic poly(propene)



syndiotactic poly(propene)



atactic poly(propene)

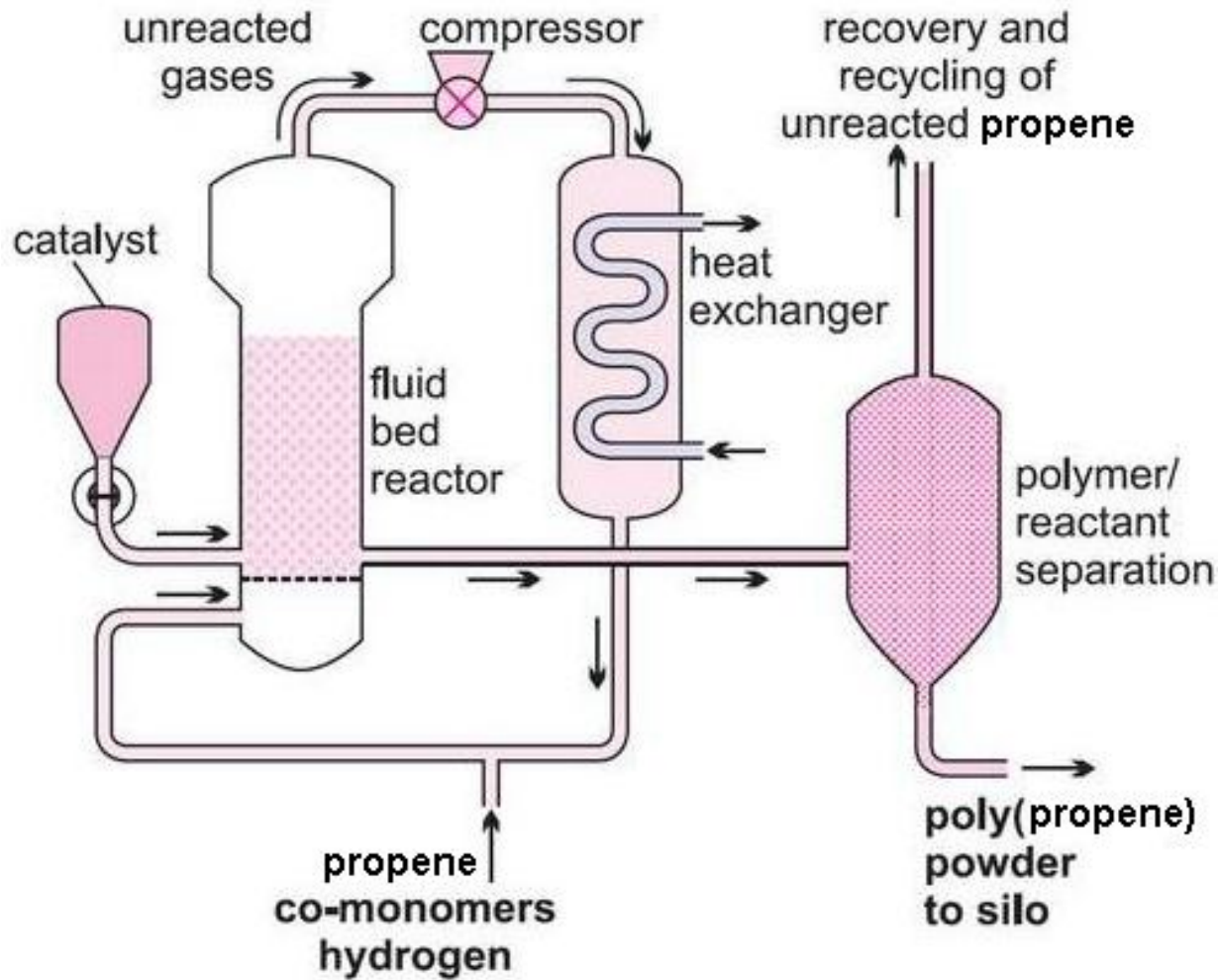
The 'one handed' structure of isotactic PP causes the molecules to form helices. This regular form permits the molecules to crystallize to a hard, relatively rigid material, which, in its pure form, melts at 440 K.

The syndiotactic polymer, because of its regular structure, is also crystalline.

Atactic chains are completely random in structure and consequently they do not crystallize.

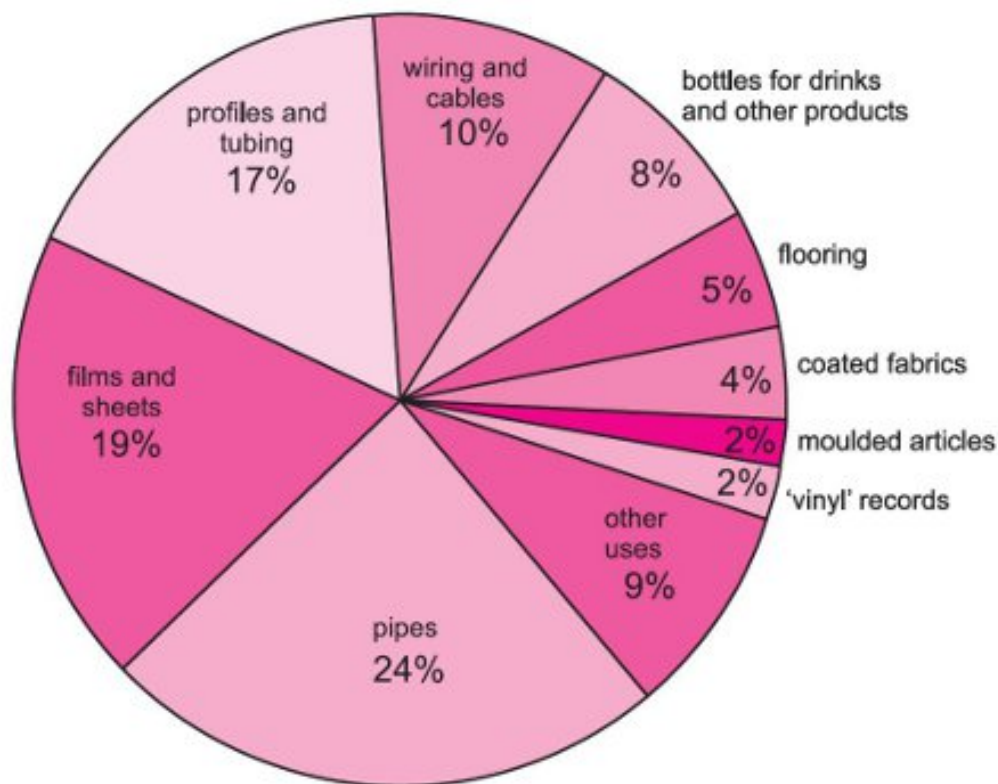
High molecular mass atactic poly(propylene) is a rubber-like material.

Commercial poly(propene) is a predominantly isotactic polymer containing 1-5% by mass of atactic material.

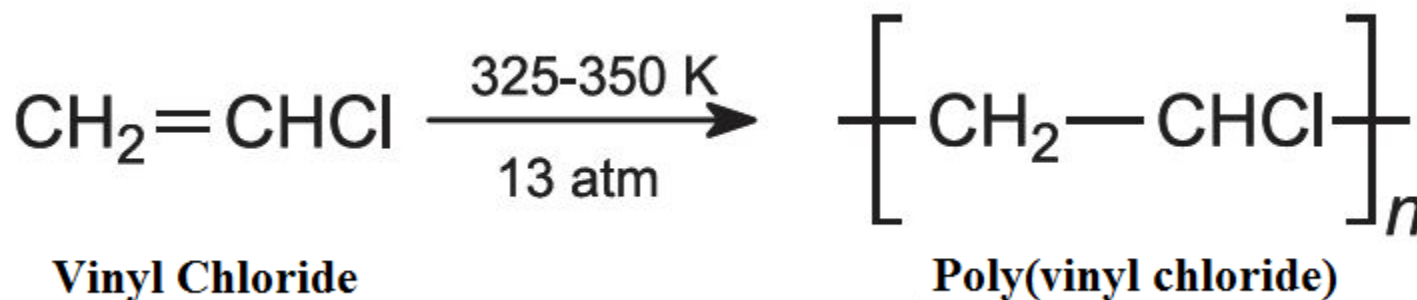


Manufacture of poly(propylene) using Low-pressure gas-phase process

Polyvinyl Chloride (PVC)



Annual Production: 62 million tonnes



In suspension polymerization an initiator, an organic peroxide is used, which is soluble in vinyl chloride. After the reaction, excess monomer is removed and the polymer is separated, by centrifuging and drying.

During polymerization, the polymer precipitates out as it is formed, since it is insoluble in the monomer. It is used for the extrusion, injection moulding and film-making processes.

It is estimated that nearly 80% of the polymer is produced by suspension polymerization.

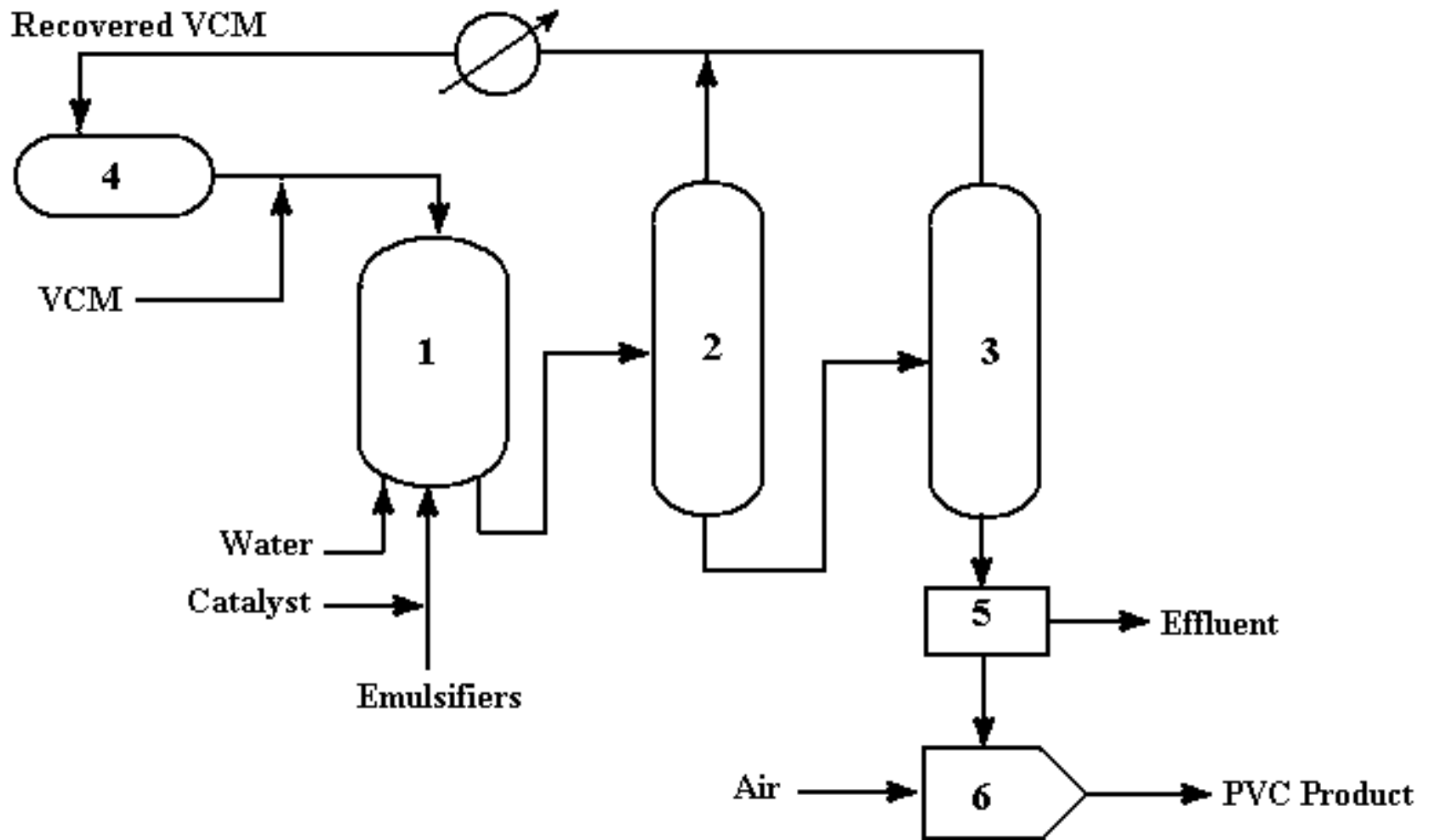
Alternatively, vinyl chloride is polymerized as an emulsion in water.

Ammonium peroxodisulfate is often used as the catalyst as it is soluble in water.

The monomer is in the form of very fine droplets rather similar to paint.

On evaporation after the polymerization, a very fine powder is formed.

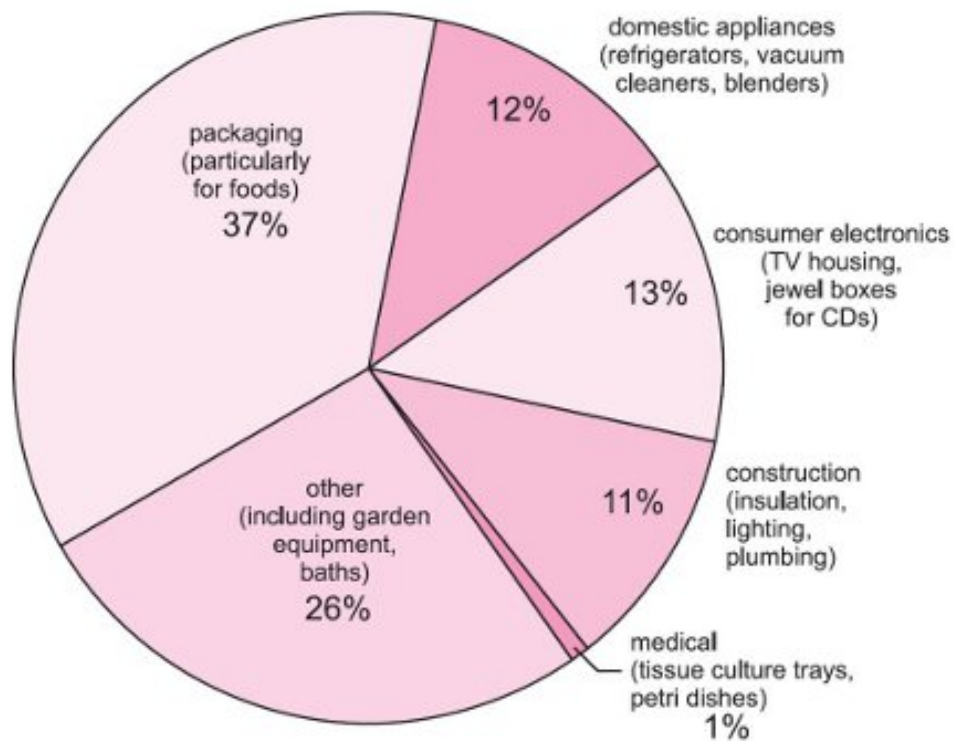
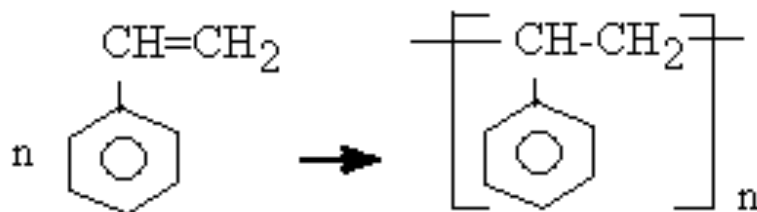
This is used as a coating on, for example, flooring and wallpaper.



1: Reactor, 2: Blowdown vessel, 3: Stripping Column, 4: VCM Storage,
5: Centrifuge, 6: Drier

Flow Sheet for the Production of Polyvinyl Chloride (PVC)

Polystyrene



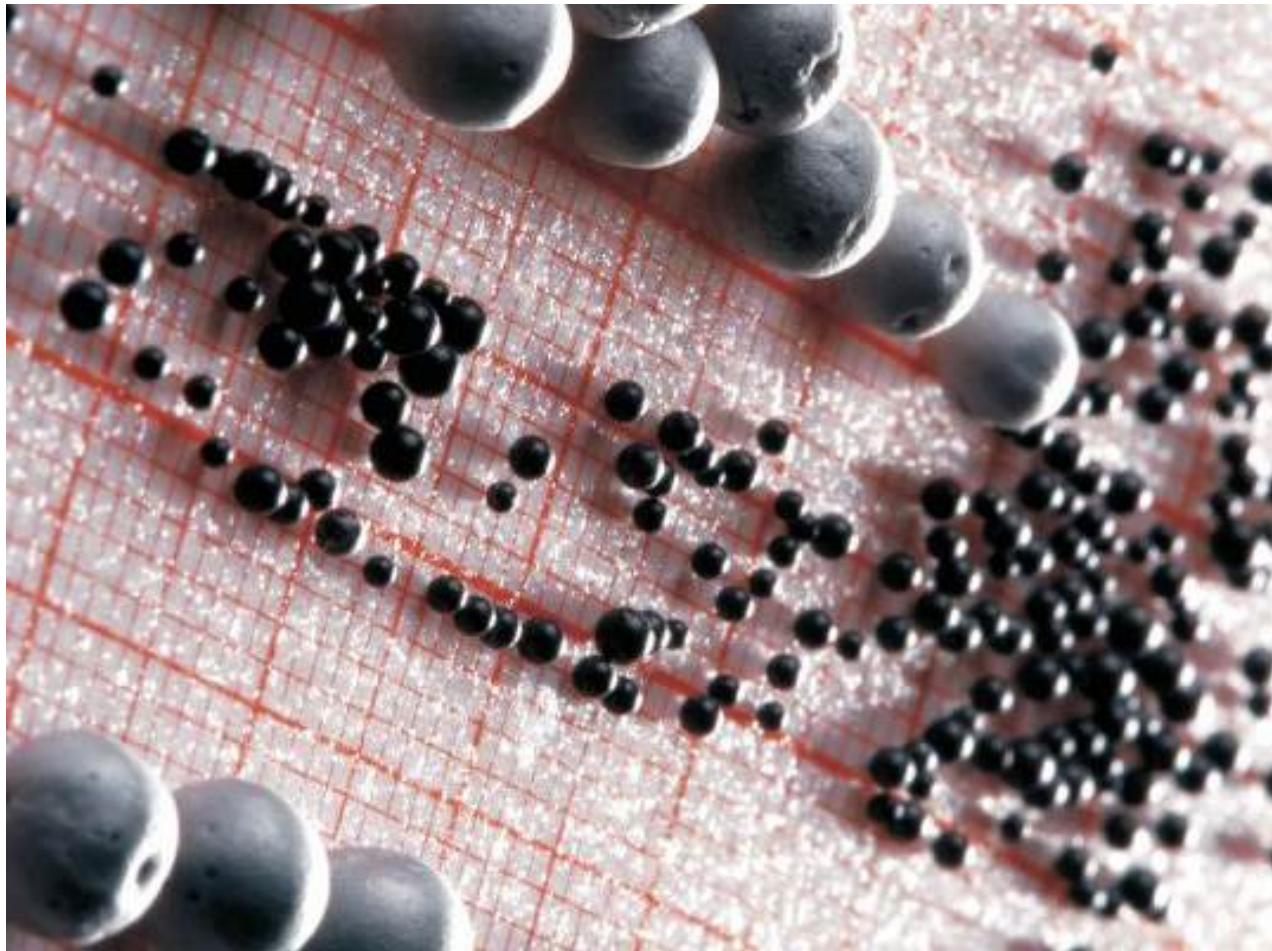
Annual Production: 27 million tonnes

Expandable Polystyrene

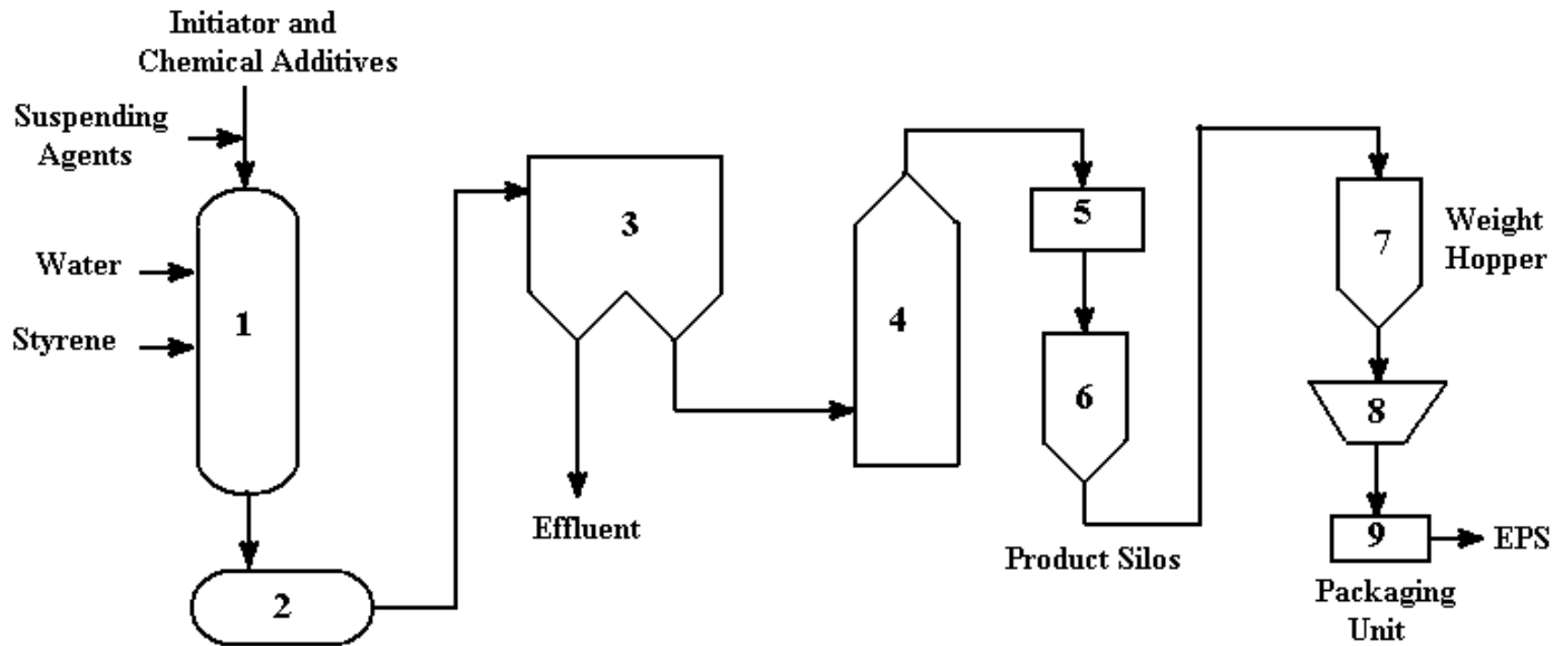
Expandable polystyrene is manufactured as beads containing pentane.

When they are heated in steam, the hydrocarbon volatilizes and the beads expand. These are subsequently blown into moulds and fused by further steaming and then cooling.

The expanded polystyrene has good thermal insulation and shock absorbing properties.

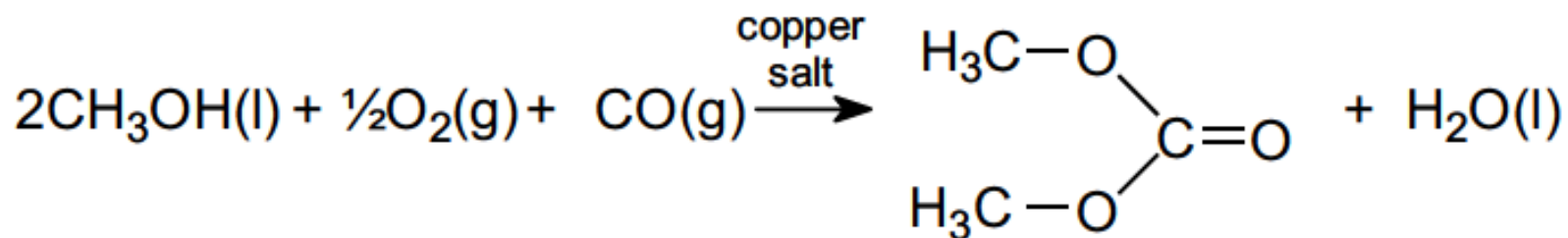


These poly(styrene) beads are shown prior to and after expansion. They were impregnated during manufacture with very fine particles of graphite to improve further their ability to absorb heat

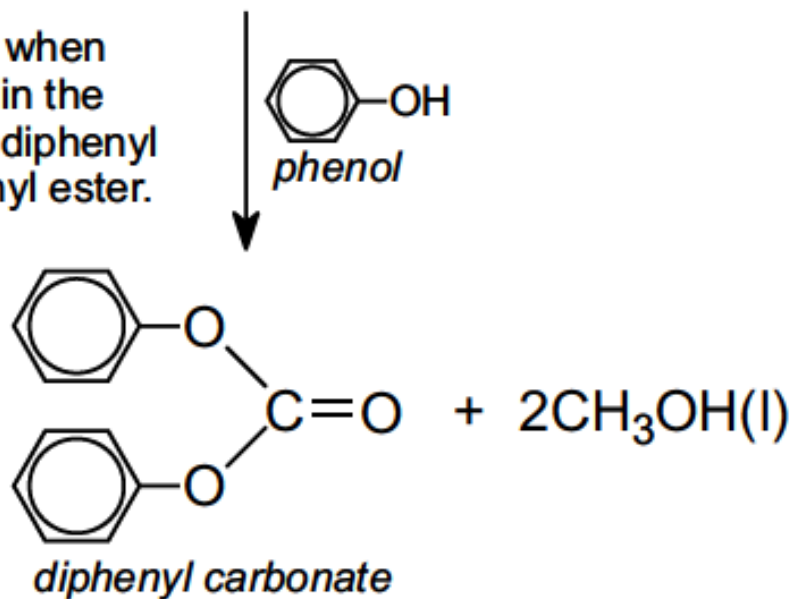


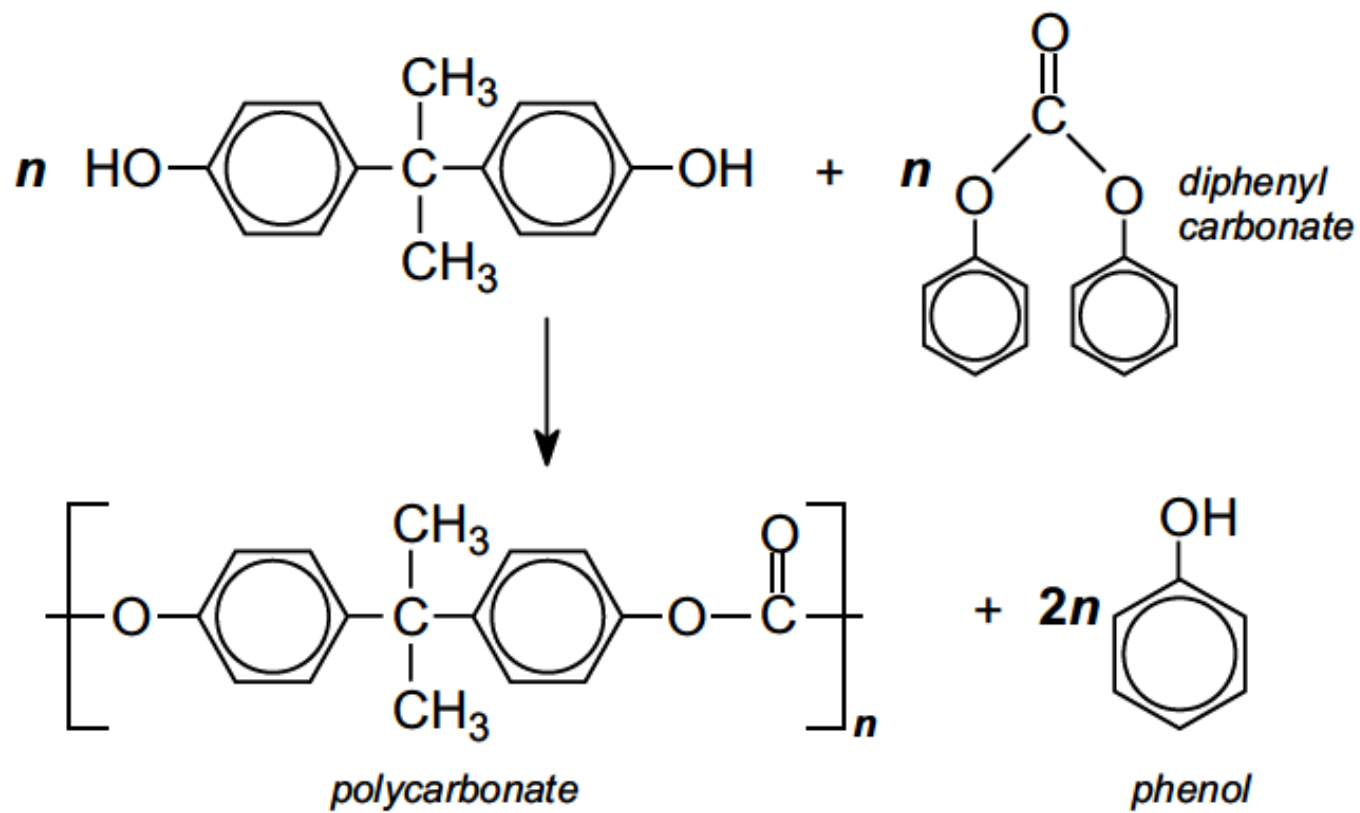
Flow Sheet for the Production of Expandable Polystyrene

Polycarbonate



The resulting ester, when heated with phenol in the liquid phase, forms diphenyl carbonate, a diphenyl ester.



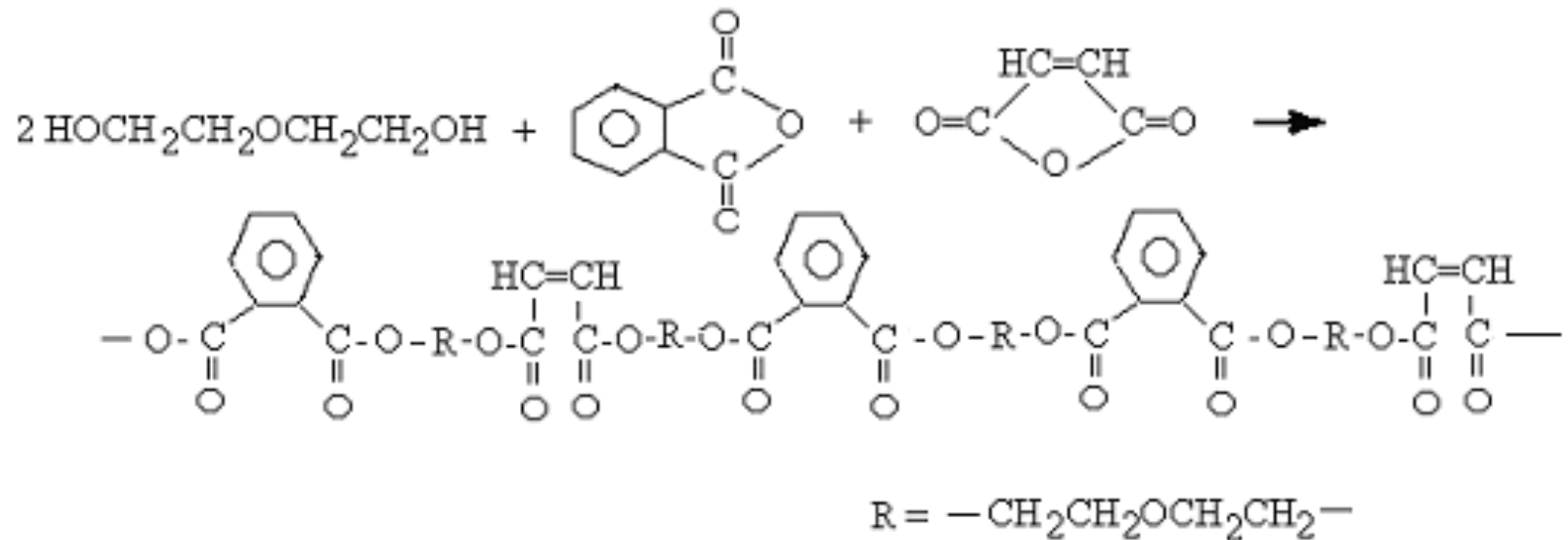


The largest application of polycarbonates is in the electrical and electronics sector for compact disks, business machine enclosures, connectors and plugs, telephones and electrical distribution devices.

The second largest application is for window glazing and related applications such as binoculars, where a virtually unbreakable molding with exceptional optical properties is required.

Polycarbonates are also blended with poly(ethylene terephthalate) or poly(butylenes terephthalate) to make impact resistant polymer alloys useful, for example, for automobile bumpers and even for side panels.

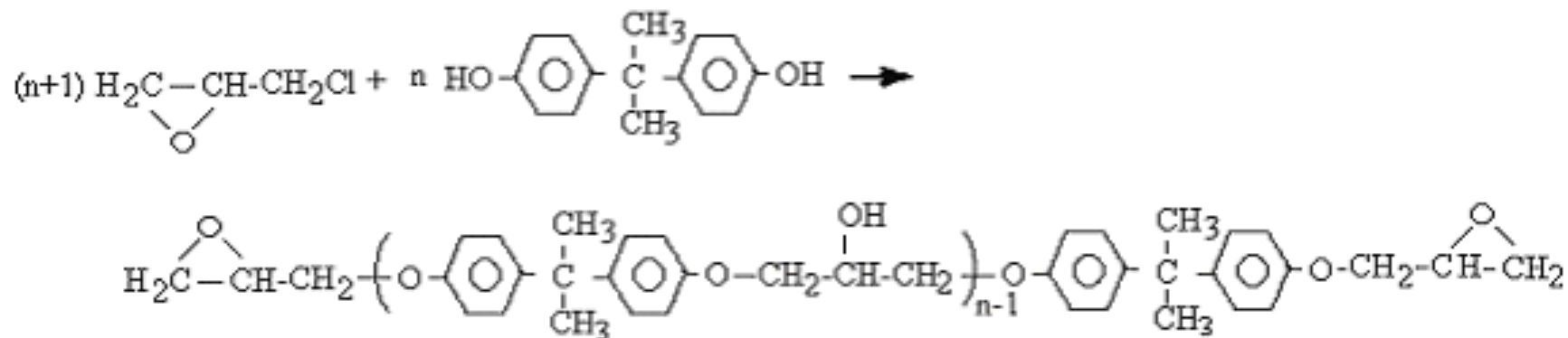
Unsaturated Polyester Resin



Unsaturated polyesters reinforced with glass cloth or fibers are used for the automobiles. They are used as a metal replacement for the fabrication of pipe and storage tanks, and in the manufacture of small boats and even large minesweepers, where the absence of metal is essential.

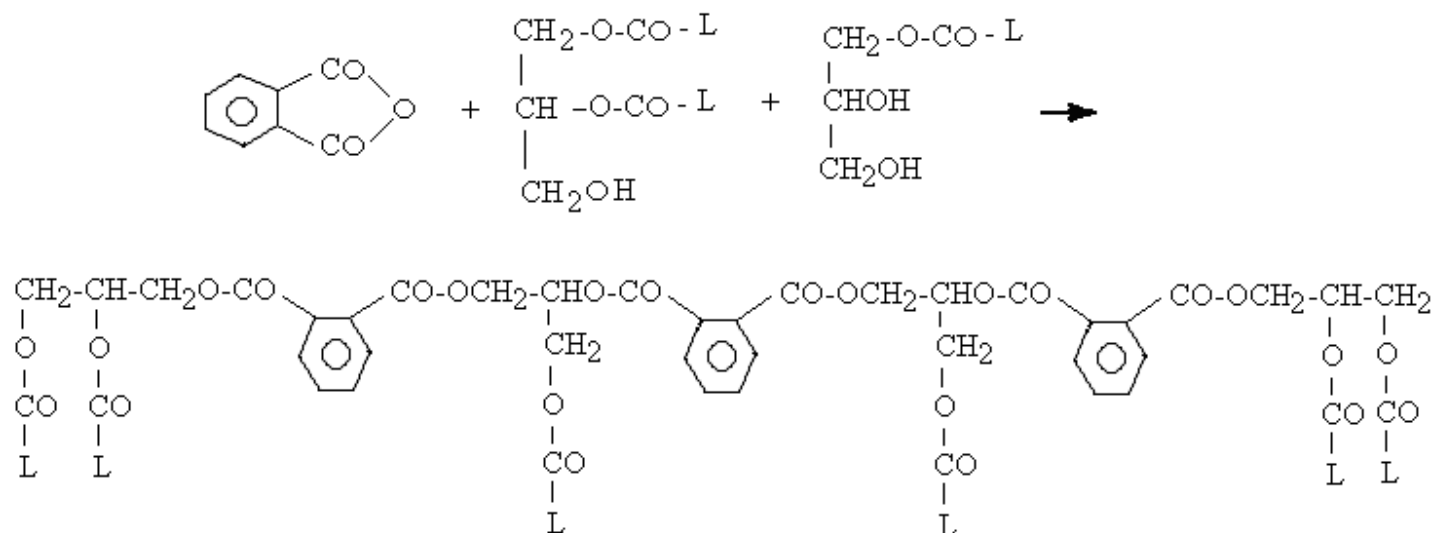
Epoxy Resin

Epoxy resins are oligomers resulting from the condensation of bisphenol A and epichlorohydrin. When cross-linked, they are hard, chemically resistant, and dimensionally stable, and have excellent electrical properties. Their largest use is for protective coatings of metal containers, appliances and ships, as well as for general maintenance where resistance to severe corrosion is required. They are used in the computer industry for “potting” electrical components. The second largest use is in fibre-reinforced composites for circuit boards, aerospace components, and sporting equipment. Other uses include adhesives, sealants, patching and flooring compounds, and castings.



Alkyd Resins

Alkyd resins are oligomers in which polyester functions have been inserted into natural “drying oils”.



The incorporation of the polyester function gives the paint film greater solvent and ultraviolet resistance and also imparts somewhat greater corrosion-resistance.

Thank you!